DISASTER MANAGEMENT

Course Material

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1. Disaster: Concept and Definition

A **disaster** is a serious disruption in the normal functioning of a community or society which leads to significant loss of life, property, environment, and economy. It occurs when the impact of a hazard exceeds the capacity of people to cope using their own resources.

Key Characteristics

- Causes large-scale damage and losses
- Disrupts normal life and daily activities
- Requires external support for recovery
- Every hazard does not necessarily become a disaster

Note: An earthquake may not cause much damage in a well-prepared city, but the same event can lead to severe destruction in an unprepared area.

2. Hazard

A **hazard** is any event or condition that has the potential to cause harm to people, property, or the environment.

Types of Hazards

1. **Natural Hazards:**
   Earthquakes, floods, cyclones, droughts, landslides
2. **Man-made Hazards:**
   Industrial accidents, fires, chemical spills, nuclear incidents

Note: Hazards cannot always be avoided, but their effects can be reduced through proper planning and preparedness.

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3. Vulnerability

Vulnerability refers to the conditions that make people, communities, or structures more likely to be affected by a hazard.

Types of Vulnerability

- **Physical:** Weak construction, unsafe locations
- **Social:** Poverty, illiteracy, age, disability
- **Economic:** Limited financial resources
- **Environmental:** Deforestation, soil erosion, climate change

Example: People living in poorly constructed houses in flood-prone areas are more vulnerable to floods.

4. Risk

Risk is the likelihood of loss or damage resulting from the interaction of hazard and vulnerability, considering the available capacity.

- Higher hazard and vulnerability increase risk
- Better capacity reduces risk

In simple terms, risk depends on how severe the hazard is, how vulnerable people are, and how prepared they are.

Example: Flooding in an area with poor drainage and no warning system leads to high risk, whereas proper planning and warning systems can reduce the risk.

5. Capacity

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Capacity refers to the strengths, resources, and abilities available within a community to manage and reduce disaster impacts.

Types of Capacity

- **Physical:** Strong buildings and infrastructure
- **Institutional:** Government bodies and disaster response teams
- **Social:** Public awareness and trained volunteers
- **Economic:** Availability of funds and resources

Note: Improving capacity is one of the most effective ways to reduce disaster risk.

6. Relationship Between Hazard, Vulnerability, Risk and Capacity

A disaster is not caused by a single factor. It results from the combined effect of hazard, vulnerability, and capacity.

$$Risk = (Hazard \times Vulnerability) / Capacity$$

- Hazard provides the potential threat
- Vulnerability increases the possible damage
- Capacity helps in reducing the impact

A disaster occurs when a hazard affects a highly vulnerable population with low capacity.

Note: Hazards are natural, but disasters occur due to vulnerability and lack of preparedness.

7. Disaster and Development

Disaster and development are closely related. The level and type of development influence how severely a disaster affects a region.

- Poor planning increases disaster risk
- Unplanned urban growth increases vulnerability
- Environmental degradation leads to frequent disasters

Examples

- Flooding due to blocked drainage systems
- Landslides due to deforestation and hill cutting

It can be said that many disasters are the result of improper or unsustainable development.

8. Disaster Management Cycle

Disaster Management involves planning, organizing, and implementing measures to reduce the impact of disasters and ensure quick recovery. Disaster management is a continuous and ongoing process.

Disaster Management Cycle

1. **Prevention** – Avoiding disaster situations

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2. **Mitigation** – Reducing the severity of impact
3. **Preparedness** – Planning and training before disaster
4. **Response** – Immediate actions during disaster
5. **Recovery** – Restoration and rehabilitation

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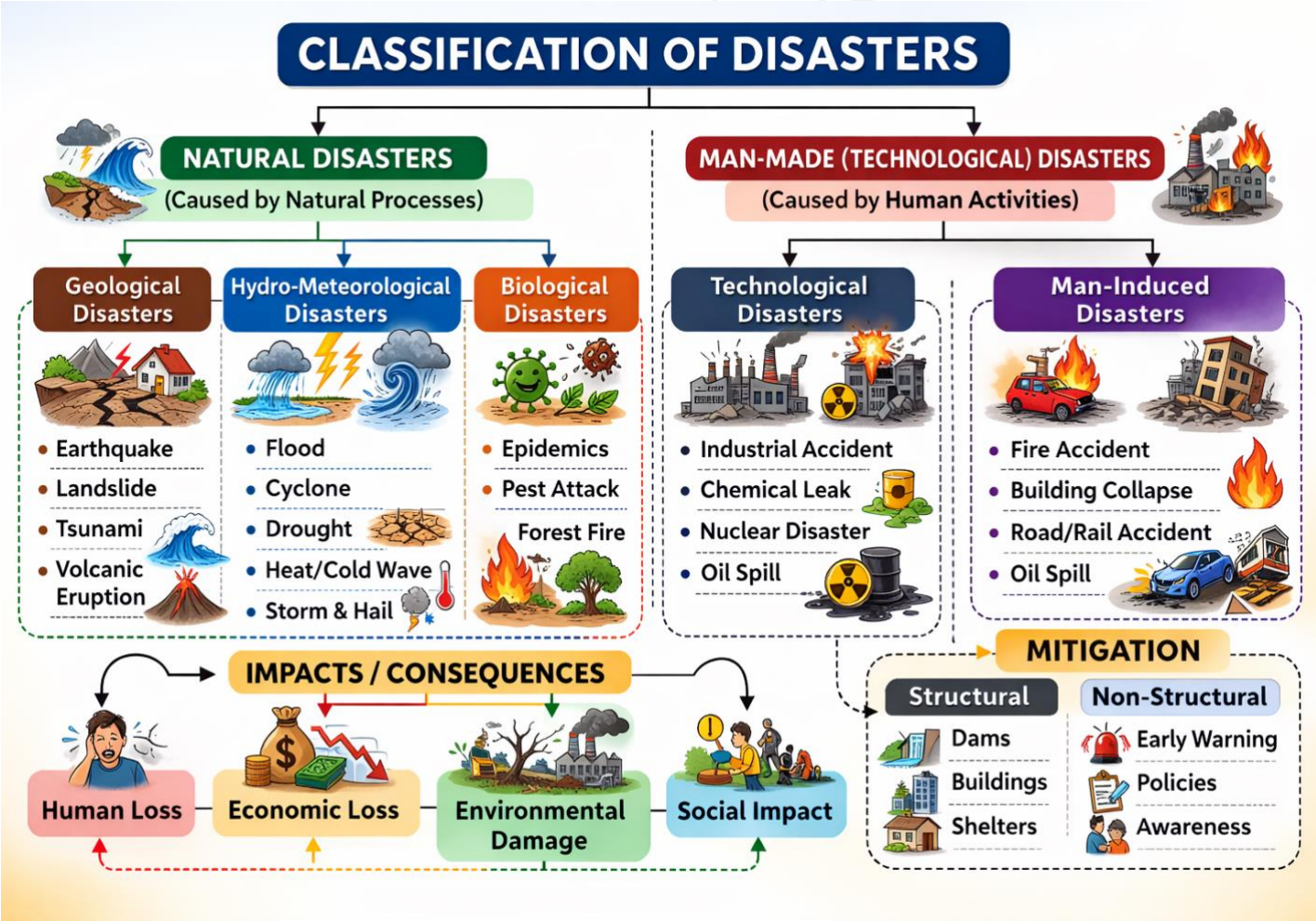
1. Introduction to Disaster

A disaster is a serious disruption in the normal functioning of a community or society which leads to widespread human, material, economic or environmental losses. It occurs when the impact of a hazardous event exceeds the ability of the affected community to cope using its own resources. Disasters may be natural or man-made, and their severity depends on factors such as vulnerability, preparedness and available resources.

2. Classification of Disasters

Disasters are broadly classified into two main categories based on their origin:

- Natural Disasters
- Man-made (Technological) Disasters
  This classification helps in understanding their causes and planning appropriate mitigation measures.



3. Natural Disasters

I. **Geological Disasters**

Geological disasters occur due to processes within the Earth's interior and are mainly associated with tectonic plate movements and instability in geological formations. These disasters are usually sudden and unpredictable in nature and can cause severe damage to life, infrastructure and the environment, especially in geologically weak regions.

Examples: Earthquakes, landslides, tsunamis and mining disasters

Causes:

• **Tectonic plate movement:** Movement of Earth's plates creates stress in the crust, which is released suddenly in the form of earthquakes and other geological disturbances.
• **Weak geological structures:** Areas with loose soil or unstable rock formations are more prone to landslides and ground failure.
• **Human activities such as mining and blasting:** Excessive excavation and use of explosives disturb the natural stability of land, increasing the chances of disasters.

II. **Hydro-Meteorological Disasters**

Hydro-meteorological disasters are related to atmospheric and water processes and are strongly influenced by climatic conditions and environmental changes. These disasters occur frequently and have a major impact on human settlements, agriculture and water resources, often leading to large-scale damage and displacement of people.

Examples:
• Floods
• Cyclones
• Droughts
• Heat waves and cold waves
• Thunderstorms and hailstorms

Causes:

- **Heavy and prolonged rainfall:** Excess rainfall over a short period leads to flooding and waterlogging in low-lying areas.
- **Climate variability and climate change:** Changes in weather patterns increase the intensity and frequency of extreme events such as cyclones and heat waves.
- **Deforestation:** Removal of vegetation reduces water absorption and increases surface runoff, leading to floods and soil erosion.
- **Poor drainage and water management systems:** Inadequate drainage systems prevent proper flow of water, causing urban flooding and water accumulation.

III. **Biological Disasters**

Biological disasters are caused by living organisms such as bacteria, viruses or pests and mainly affect public health, agriculture and ecosystems. These disasters can spread rapidly

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over large areas if not properly controlled and may result in serious health issues and economic losses.

Examples: Epidemics (COVID-19, dengue), Pest attacks, Forest fires

Causes:

- **Poor sanitation and hygiene:** Unhygienic conditions promote the growth and spread of disease-causing organisms.
- **Climate change:** Changes in temperature and rainfall create favourable conditions for the spread of diseases and pests.
- **Increased human-animal interaction:** Close contact between humans and animals increases the chances of transmission of infectious diseases.

4. Man-Made (Technological) Disasters

Technological Disasters

Technological disasters occur due to the failure or breakdown of industrial and technological systems. These disasters are generally associated with the use of hazardous materials, complex machinery and advanced technology. They can lead to serious damage to human life, property and the environment, and often have long-term consequences such as pollution and health hazards. Such disasters usually require immediate technical response and strict control measures.

Examples:

- **Chemical disasters:** Leakage or release of toxic chemicals or gases from industries, which can cause poisoning, breathing problems and contamination of air, water and soil.
- **Industrial accidents:** Explosions, fires or machinery failures in factories that can injure workers and damage surrounding areas.
- **Nuclear disasters:** Release of radioactive substances from nuclear power plants, leading to long-term health issues such as cancer and environmental degradation.

Causes:

- **Equipment failure:** Malfunctioning or breakdown of machines due to defects, aging or improper use.
- **Human negligence:** Carelessness, lack of training or mistakes during operation and handling of equipment.
- **Poor safety standards and maintenance:** Failure to follow safety procedures, lack of regular inspection and improper maintenance of systems.

IV. **Man-Induced Disasters**

Man-induced disasters are directly caused by human activities and are mainly the result of negligence, lack of awareness, improper planning or violation of safety rules. These disasters are generally preventable if proper precautions, regulations and safety measures are followed. They mostly occur in everyday life and affect both urban and rural areas.

Examples:

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- **Building collapse:** Failure of structures due to poor design, use of low-quality materials or overloading beyond capacity.
- **Fire accidents:** Fires caused by electrical short circuits, unsafe storage of flammable materials or human carelessness.
- **Road and rail accidents:** Accidents caused due to human error, over-speeding, poor maintenance of vehicles or violation of traffic rules.
- **Oil spills:** Leakage of oil into water bodies during transport or industrial activities, leading to severe environmental damage and harm to marine life.

Causes:

- **Poor construction practices:** Lack of proper engineering design and use of substandard materials in construction.
- **Overloading of structures:** Applying loads beyond the safe capacity of buildings or systems.
- **Ignoring safety regulations:** Failure to follow safety rules, guidelines and standard procedures during construction and operations.

5. Causes of Disasters

Disasters occur due to a combination of natural, human and technological factors. In many situations, natural hazards become disasters because of human negligence and lack of preparedness.

Major Causes:

- Natural causes: earthquakes, heavy rainfall, droughts, climate variations
- Human causes: unplanned urbanization, deforestation, poor infrastructure
- Technological causes: equipment failure, lack of maintenance, inadequate training

6. Consequences / Impacts of Disasters

Disasters have wide-ranging effects on society, economy and environment. Their impacts can be immediate as well as long-term.

a) **Human Impacts:** Disasters directly affect human life and health.

- **Loss of life:** Many people may die due to collapse of buildings, floods, fires or other hazards.
- **Injuries and diseases:** Survivors may suffer from injuries, infections and spread of diseases due to poor sanitation after disasters.
- **Mental stress and trauma:** People may face fear, anxiety and emotional distress due to loss of family members, homes and security.

b) **Economic Impacts:** Disasters cause serious financial losses to individuals and the nation.

- **Damage to property and infrastructure:** Houses, roads, bridges and public utilities are damaged or destroyed.
- **Loss of livelihood:** People lose jobs, crops, businesses and sources of income.
- **Increased expenditure on recovery and rehabilitation:** Governments and individuals have to spend large amounts on rebuilding and relief activities.

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c) **Environmental Impacts:** Disasters disturb the natural environment and ecosystems.

- **Soil erosion:** Floods and landslides wash away fertile soil.
- **Water pollution:** Contamination of water bodies occurs due to debris, chemicals or waste.
- **Loss of biodiversity and natural resources:** Plants, animals and ecosystems are damaged or destroyed.

d) **Social Impacts:** Disasters affect the social structure and normal life of communities.

- **Migration and displacement:** People are forced to leave their homes and move to safer places.
- **Disruption of education and services:** Schools, hospitals and other services stop functioning temporarily.
- **Breakdown of community structure:** Social relationships and support systems are disturbed.

7. Control and Mitigation of Disasters

Although disasters cannot be completely prevented, their impacts can be minimized through proper planning, preparedness and management. Effective mitigation helps in reducing vulnerability and improving resilience.

a) Structural Measures

Structural measures refer to physical or engineering constructions used to reduce the impact of disasters. These measures are designed to make buildings and infrastructure strong enough to withstand hazards.

- **Earthquake-resistant buildings:** Structures are designed using special techniques and materials so that they can resist seismic forces and reduce damage during earthquakes.
- **Flood control dams and drainage systems:** Dams help in controlling excess water, while proper drainage systems ensure smooth flow of rainwater, reducing the chances of flooding.
- **Cyclone shelters:** These are specially constructed safe buildings where people can take shelter during cyclones, protecting them from strong winds and heavy rainfall.

b) Non-Structural Measures

Non-structural measures do not involve physical construction but focus on planning, policies and awareness to reduce disaster risk.

- **Land use planning:** Proper planning of land helps avoid construction in high-risk areas such as flood-prone or landslide-prone zones.
- **Early warning systems:** These systems provide advance information about disasters, allowing people to take timely action and reduce losses.
- **Public awareness and disaster education:** Educating people about safety measures, preparedness and response helps in minimizing damage during disasters.
- **Implementation of policies and building codes:** Government rules and regulations ensure safe construction practices and disaster preparedness at all levels.

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8. Global Disaster Trends

In recent years, disasters have increased both in number and intensity. This is mainly due to population growth, rapid urbanization and climate change. As a result, disasters are causing more damage to life and property, especially in developing countries. Cities are becoming more vulnerable because of high population density, poor planning and inadequate infrastructure.

Observations:

a) **Rising Frequency of Disasters**-The number of disasters across the world has increased over the last few decades due to changes in climate and increasing human activities.
b) **Increased Human and Economic Losses**-Disasters now result in greater loss of life and higher economic damage, particularly in countries with limited resources.
c) **Urban Vulnerability**-Urban areas are more at risk because of overcrowding, poor drainage systems and unplanned development.
d) **Climate Change Link**-Climate change is leading to more extreme weather events, making disasters more severe and unpredictable.

Example: Cyclone Idai (2019, Mozambique) caused heavy flooding, large-scale displacement and health issues.

9. Emerging Risks of Disasters

In today's scenario, new types of risks are emerging which are increasing the chances and impacts of disasters.

a) Climate Change

Climate change refers to long-term changes in temperature and weather patterns.
Impacts include:

- Increase in cyclones, floods and heatwaves
- Rise in sea level leading to coastal flooding
- Irregular rainfall causing droughts or floods

b) Urban Disasters

Urban disasters are common in cities where infrastructure is not properly developed.
Causes include:

- Unplanned urban growth
- Poor drainage and waste management
- Overcrowded housing
  Examples: Mumbai floods, fire accidents in high-rise buildings

c) Technological / Industrial Disasters

These disasters occur due to industrial activities or failure of technology.
Examples include:

- Chemical leaks such as Bhopal Gas Tragedy
- Nuclear accidents
- Transport accidents involving hazardous materials

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d) Pandemics
A pandemic is the spread of an infectious disease across countries or continents.
Examples: COVID-19, Spanish Flu
These situations show the weakness of health systems and economic structure.

e) Compound Disasters
These occur when one disaster leads to another or when multiple disasters happen together.
Example: Earthquake followed by tsunami and then industrial accident.

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